SH Loss Calibration: Before / After

`shg_mode_survey · loaded_sweep1 + trial_grid + wide_window`

413 refined crossings across 26 geometries, with a flat, empirically-calibrated 100 dB/cm SH-band absorption added to the existing loss budget (geometry.SH_ABSORPTION_CALIBRATED_DB_PER_CM, confinement-weighted -- see build_assumptions_report.py section 3b for the full derivation). AlN's own measured k_e (AM187-LT6) was explored first but REJECTED as the SH-loss assumption -- it implied ~100,000 dB/cm, checked against real device output and found 3-4 orders of magnitude too high (kept in the data/notes for context, not used here).

NCE (eta_pct_per_W_cm2) does not depend on loss at all -- it is EXACTLY unchanged. Only loss-limited PEAK efficiency can move, and only via SH loss specifically -- the calibrated absorption is SH-band only, pump loss is untouched.

PEAK EFFICIENCY DROP FACTOR -- RIDGE FAMILY

1.6x median, 10.8x max

bare AlN ridge had ~zero SH-side loss before -- hit hardest

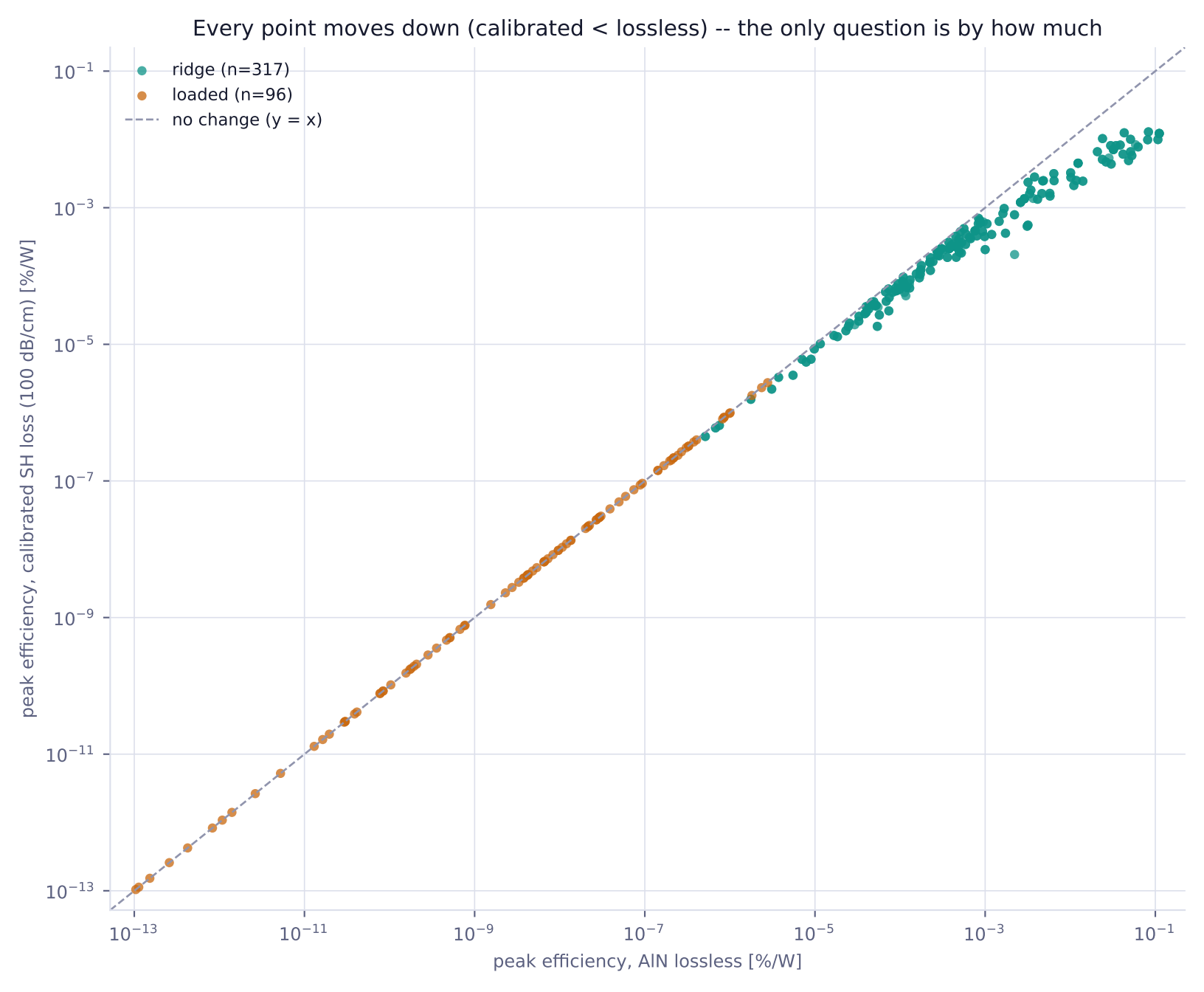
PEAK EFFICIENCY DROP FACTOR -- LOADED FAMILY

1.0x median, 1.0x max

already had a lossy TiO2 strip -- added absorption is a smaller relative penalty

CAUTION

100 dB/cm is a round working number inside an empirically-bounded range (15-680 dB/cm depending on an unmeasured input-coupling efficiency), not a measurement of any specific physical mechanism. pedestal family excluded -- no field exports available (unrelated to this calibration, which needs none; see this file's module docstring).


Every point moves down (calibrated < lossless) -- the only question is by how much
ridge (n=317)
loaded (n=96)
no change (y = x)
peak efficiency, calibrated SH loss (100 dB/cm) [%/W]
peak efficiency, AlN lossless [%/W]
10^{-1}
10^{-3}
10^{-5}
10^{-7}
10^{-9}
10^{-11}
10^{-13}

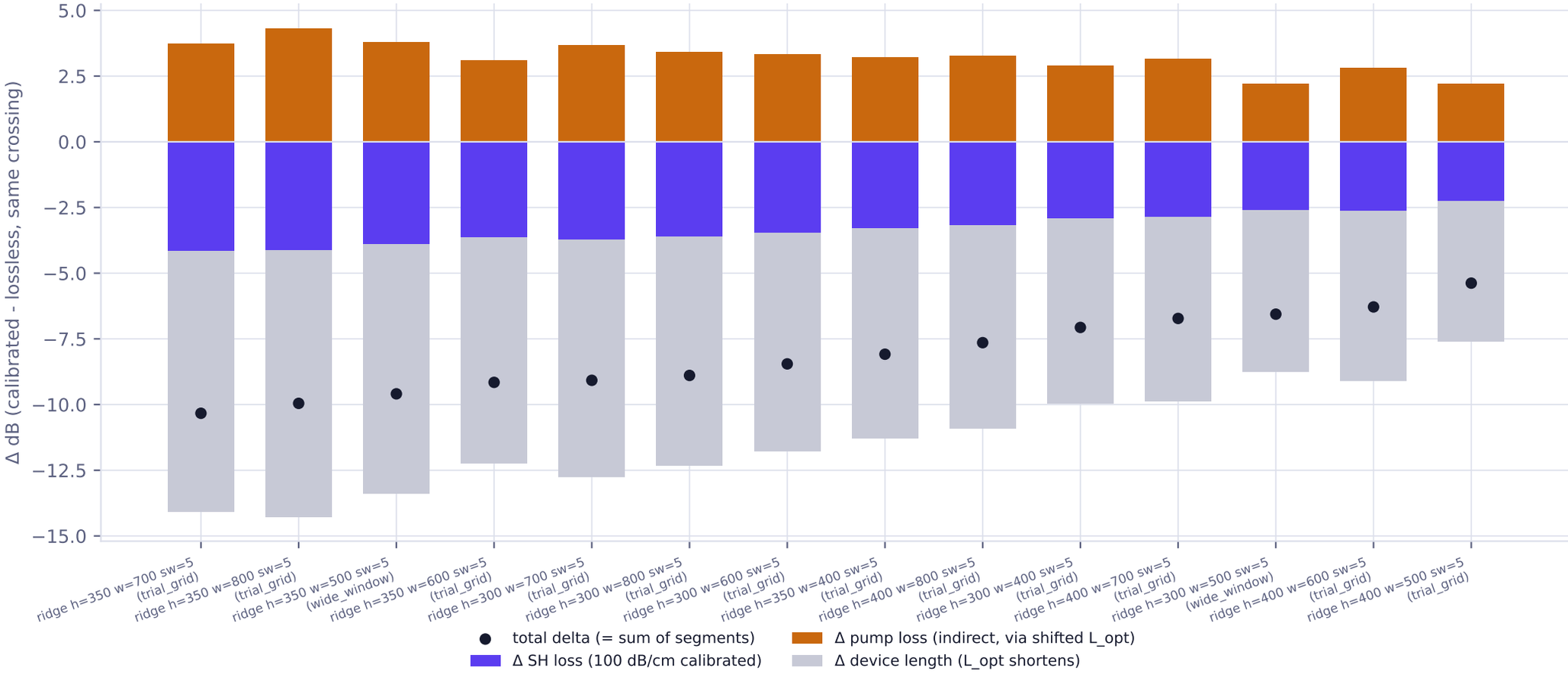
Top 15 crossings by lossless peak efficiency -- where do they rank with calibrated SH loss?

run · geometry · pair

#	GEOMETRY	PUMP→SH	LOSSLESS [%/W]	CALIBRATED [%/W]	NEW RANK
1	ridge h=350 w=500 sw=5 (wide_window)	TM00→TM04	0.111	0.0122	#7
2	ridge h=350 w=500 sw=5 (wide_window)	TM00→TM04	0.111	0.0122	#8
3	ridge h=350 w=500 sw=5 (trial_grid)	TM00→TM04	0.111	0.0122	#5
4	ridge h=350 w=500 sw=5 (trial_grid)	TM00→TM04	0.111	0.0122	#6
5	ridge h=350 w=700 sw=5 (trial_grid)	TM00→TM04	0.107	0.00994	#13
6	ridge h=350 w=700 sw=5 (trial_grid)	TM00→TM04	0.107	0.00994	#14
7	ridge h=350 w=400 sw=5 (trial_grid)	TM00→TM04	0.083	0.0129	#1
8	ridge h=350 w=400 sw=5 (trial_grid)	TM00→TM04	0.083	0.0129	#2
9	ridge h=350 w=600 sw=5 (trial_grid)	TM00→TM04	0.0813	0.00988	#15
10	ridge h=350 w=600 sw=5 (trial_grid)	TM00→TM04	0.0813	0.00988	**#16 ↓**
11	ridge h=300 w=700 sw=5 (trial_grid)	TM00→TM04	0.0628	0.00777	**#24 ↓**
12	ridge h=300 w=700 sw=5 (trial_grid)	TM00→TM04	0.0628	0.00777	**#25 ↓**
13	ridge h=300 w=600 sw=5 (trial_grid)	TM00→TM04	0.0586	0.00836	**#17 ↓**
14	ridge h=350 w=700 sw=5 (trial_grid)	TM10→TM04	0.0529	0.00579	**#36 ↓**
15	ridge h=350 w=700 sw=5 (trial_grid)	TM10→TM04	0.0529	0.00579	**#37 ↓**

6 of the original top 15 fall out of the top 15 once the calibrated SH loss is included (orange, marked with new rank).

Why peak efficiency drops: exact dB delta per geometry (worst first)



Each bar is the SAME crossing's best-by-lossless-peak-efficiency result, decomposed twice (nce_decomposition.decompose, include_sh_calibrated=False/True) -- shape_dB/ field_dB are identical either way (depend only on overlap_shape and eta_pct_per_W_cm2, neither of which depends on loss). alpha_p itself is never touched by the calibrated absorption, but pump_loss_dB still shifts (small, usually positive) because it's evaluated at L_opt, which shortens once SH loss rises -- not a separate mechanism, just the same L_opt plugged into the pump term. The black dot is the exact sum of the three segments, verified against the direct peak-efficiency ratio.

Reality check: what SH loss is consistent with measured device output?

`ridge h=350 w=500 nm, TM00→TM04 -- closest match to the measured device`

Given a real device's actual, fixed propagation length (not an optimized L_opt) and actual pump power reaching the waveguide, shg_physics.loss_limited_length's own model (P_SH = eta_SI * P_pump^2 * L_eff(L; alpha_p, alpha_s)^2) can be inverted for the SH-band loss that reproduces a measured SH output -- independent of whether any particular ellipsometry sample's k is trustworthy. Using this geometry's own simulated eta_SI and currently-modeled native pump loss (15.8 dB/cm, no AlN):

PUMP POWER (IN WAVEGUIDE)	INTERACTION LENGTH	TARGET SH OUTPUT	IMPLIED alpha_SH BOUND
100 mW	2000 um (2 mm)	20 nW	**<= 679 dB/cm**
100 mW	2000 um (2 mm)	200 nW	**<= 235 dB/cm**

**AM187-LT6 k_e implies 100,161 dB/cm for this same crossing --
147x to 427x above what the device's own measured output permits.**

**The 100 dB/cm calibrated value gives 101.1 dB/cm total SH loss here
(scattering + calibrated absorption). Predicted SH output at 10/25/50% coupling:
44 / 277 / 1107 nW -- closely tracks the "10s of nW typically"
benchtop population at low coupling, within ~1.4x of the 200nW best case at 25%
coupling. At the pessimistic 50% coupling used in the table above it over-predicts
(1107 vs. 200 nW) -- points to true coupling being below
50% for that device, not to 100 dB/cm being too low.**

20 nW is the user's suggested representative floor (most devices clear at least this much; 200 nW was closer to the best-observed case, shown for context). The alpha_SH bound table above is fixed at 100 mW (50% of 200 mW incident, the optimistic end of the 10-50% coupling-efficiency uncertainty) since that was specifically requested for this reality check; it is a bound, not a measurement of alpha_SH itself -- it assumes the modeled eta_SI and native pump loss for this geometry are correct, and that the real device isn't compensating for higher SH loss some other way (e.g. a shorter effective interaction length than the nominal 2 mm, imperfect mode matching reducing the EFFECTIVE eta).

Notes & provenance

Data: runs/loaded_sweep1/crossings.csv, runs/trial_grid/crossings.csv, runs/wide_window/crossings.csv, refined rows with a reprocessed AlN-loss pair only (aln_pump_loss_dB_per_m not null -- a leftover filter from the AM187 investigation; the calibrated scenario itself needs only sh_power_in_core, present on every refined row regardless). Generated via build_aln_loss_report.py / nce_decomposition.py.

CALIBRATED SCENARIO (used throughout this report): geometry.sh_absorption_calibrated_dB_per_m -- a flat 100 dB/cm, confinement-weighted by the mode's power fraction in geom.core_mask() (sh_power_in_core), SH-band only, never added to pump loss. Empirically calibrated against real fabricated-device output, 2026-10-12 -- see build_assumptions_report.py section 3b and PLAN.md for the full derivation. Computed entirely from already-stored columns, no field re-export or EMode re-solve needed.

LOSSY SCENARIO (AM187-LT6 k_e, shown in notes/title context only, NOT the headline comparison): geometry.aln_core_loss_dB_per_m(wavelength_nm), a Beer-Lambert conversion of the AM187-LT6-measured k_e (extraordinary axis, 3-Lorentzian fit -- materials/fit_aln_measured_k.py), same confinement-weighting pattern. REJECTED as the SH-loss assumption 2026-10-12 -- its SH-band implication (~100,000 dB/cm) was checked against real device output and found 3-4 orders of magnitude too high (see the reality-check page). k_e only, not field-direction-resolved against k_o (k_e is ~90x k_o and is the axis d33 couples -- Py = d33*Ey^2).

Original AlN-lossless columns (pump_absorption_dB_per_m / sh_absorption_dB_per_m / peak_efficiency_pct_per_W) are untouched in crossings.csv; both the calibrated and lossy figures here are computed downstream, on the fly, from nce_decomposition.decompose.

Excluded: pedestal1/pedestal_sweep1/pedestal_vertical1 (zero field exports on either machine; re-solving from scratch was estimated at ~9.5 hours of EMode time for a geometry already being superseded by the corrected embedded 'hybrid' family -- not worth it, per 2026-10-12 decision). trial_grid is missing 11/280 refined crossings' exports even after merging the laptop's local runs folder -- those rows are excluded here too, not zero-filled.